

ARASH PARSA

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ABOUT ME

Computational biologist with a background in bioengineering, machine learning, and wet lab molecular biology. Experienced in building bioinformatics pipelines, analyzing omics datasets, and applying statistical and ML methods to biological data. Particularly interested in spatial transcriptomics, multi-omics integration, and developing computational tools that bridge bench science and data-driven discovery.

EDUCATION

Georgia Institute of Technology

Masters of Science in Data Science/Analytics

Atlanta, GA August 2022 – July 2025

University of California, Santa Cruz

Bachelors of Science in Bioengineering/Bioinformatics

Santa Cruz, CA August 2015 – June 2021

EXPERIENCE

Machine Learning Research Collaborator – Cost-Aware Clinical Phenotyping

Dr. Vivek Rudrapatna's Clinical Data Science Group, UCSF

2026 – Present

San Francisco, CA

- Initiated the collaboration independently; scoped the cost bottleneck in EHR-based phenotyping directly with the physician-scientist stakeholder and proposed a confidence-gated cascade as the solution
- Architected a four-stage pipeline: (1) *Extract* – CPU BioBERT disease-NER; (2) *Retrieve* – entity-guided context selection; (3) *Cascade* – accept high-confidence entities on the cheap path, route low-confidence and zero-hit notes to a frontier LLM; (4) *Evaluate* – mention-level precision/recall/F1 against gold annotations with a token-cost proxy
- On an early pilot over the NCBI Disease corpus, escalating ~35% of notes raised precision from 0.62 to 0.75 while consuming only ~24% of the tokens a full-note frontier-LLM approach would require
- Engineered the system to run CPU-only with no GPU dependency – a deliberate constraint making the approach deployable inside hospital firewalls where GPUs are unavailable
- Authored a literature review and an 11-slide novelty case establishing that individual cost levers (distillation, retrieval, cascading) are each proven, but none are combined into one confidence-gated cascade benchmarked across many phenotypes
- Obtained PhysioNet credentialing for MIMIC-III/IV; extending toward a weighted multi-dataset benchmark (MIMIC, n2c2) comparing distilled open-weight models against frontier models on joint accuracy-cost frontiers, with UCSF corpus validation and publication planned

Knowledge and Skills: Clinical NLP; named entity recognition; BioBERT; transformer fine-tuning; knowledge distillation; large language models; confidence-gated model cascades; benchmark design; precision/recall/F1 evaluation; cost-accuracy tradeoff analysis; MIMIC-III/IV; n2c2; NCBI Disease; credentialed clinical data; CPU-constrained inference; PyTorch; HuggingFace Transformers; Python; Git; independent research scoping

Data Science Research Assistant in Clinical Research

In collaboration with Stanford-affiliated researchers

January 2026 – Present

San Francisco, CA

- Trichoscopic Oculomics – built deep learning classification pipeline integrating spatial transcriptomics and metabolic epidemiology features to reframe androgenetic alopecia as a systemic screening opportunity
- Endometriosis Severity Prediction – developed supervised ML models (random forest, gradient boosting, logistic regression) to phenotype and preoperatively predict endometriosis severity from clinical and demographic features extracted from EHR-derived datasets
- Performed end-to-end clinical data preprocessing: missing data imputation, feature encoding, class imbalance correction (SMOTE), and cross-validated model evaluation; presented results to clinical research stakeholders

Knowledge and Skills: Machine learning; spatial transcriptomics; deep learning; clinical data analysis; EHR data; phenotyping; predictive modeling; Python; scikit-learn; TensorFlow; Pandas; cross-validation; precision medicine

Machine Learning & Data Science Research Assistant

Washington Lab, UCSF

December 2025 – March 2026

San Francisco, CA

- Systematic review study: cataloged model architectures (CNNs, LSTMs, transformers), dataset characteristics, and performance benchmarks across 50+ studies on home video-based ML classification of autism and ADHD in pediatric populations
- Developed Python scripts to extract, standardize, and visualize study metrics (sensitivity, specificity, AUC) across heterogeneous reporting formats, enabling quantitative cross-study comparison
- Manuscript to be submitted for publication in the near future

Knowledge and Skills: Systematic literature review; data analytics; AI/ML in healthcare; performance benchmarking; Python; Pandas; Matplotlib; statistical analysis; data visualization

Machine Learning Scientist

Optelligence

September 2022 – Present Berkeley, CA

- Designed and trained supervised regression and classification models (XGBoost, neural networks) for performance prediction, cutting physical testing cycles by 50%
- Built end-to-end data preprocessing pipelines in Python (Pandas, NumPy, scikit-learn) including feature engineering, outlier detection, and normalization across heterogeneous datasets
- Applied CNNs and graph neural networks in TensorFlow and PyTorch to extract latent structure from high-dimensional experimental measurements

Knowledge and Skills: Python; TensorFlow; PyTorch; scikit-learn; XGBoost; Pandas; NumPy; feature engineering; statistical analysis; model training; hyperparameter tuning; CNNs; graph neural networks

R & D Engineer

Amgen & Thermofisher

August 2021 – May 2022 Thousand Oaks, CA & Pleasanton, CA

- Automated quality control workflows using Python, reducing manual review time and maintaining 99.8% assay accuracy across 100+ daily COVID-19 diagnostic samples
- Developed statistical process control (SPC) dashboards to monitor HPLC/UPLC instrument drift, enabling early anomaly detection and reducing reagent waste by \$50K annually
- Performed multivariate data analysis on chromatography outputs using R and Excel; identified systematic variance sources and optimized parameters through iterative experimentation

Knowledge and Skills: Python; R; Excel; statistical process control; HPLC/UPLC; quality control automation; chromatography data analysis; multivariate analysis; anomaly detection

Research Assistant – Progesterone Biosynthesis (iGEM)

University of California, Santa Cruz

2018 Santa Cruz, CA

- Engineered *Y. lipolytica* for progesterone biosynthesis via metabolic pathway engineering; designed 5-gene cassette for heterologous expression
- Executed 3 parallel cloning strategies: Gibson Assembly, yeast-mediated cloning, and Cre-Lox recombination; validated constructs with gel electrophoresis, colony PCR, and Sanger sequencing
- Led international outreach across 15+ countries; co-authored 60-page published research report

Knowledge and Skills: Metabolic pathway engineering; gene cassette design; Gibson Assembly; yeast transformation; Cre-Lox recombination; plasmid cloning; CRISPR; recombinant DNA; PCR; gel electrophoresis; Sanger sequencing; primer design

Scientific Researcher & Research Consultant

Apex Energetics

January 2016 – November 2019 Irvine, CA

- Designed and maintained a structured literature database of 200+ papers/month on 15+ nutraceutical compounds; applied NLP-assisted tagging to surface trends in clinical efficacy data
- Synthesized evidence across heterogeneous study designs (RCTs, meta-analyses, in vitro) to produce quantitative summary reports, saving 20 hours/week of manual review for research leads

Knowledge and Skills: NLP-assisted literature review; evidence synthesis; RCTs; meta-analyses; clinical efficacy data; quantitative reporting; database design and maintenance

PROJECTS

Spatial Transcriptomics Analysis of Human Brain Tissue

Personal Project – GitHub

2025 – 2026

Berkeley, CA

- Analyzed 10x Visium spatial transcriptomics data from human dorsolateral prefrontal cortex (spatialLIBD dataset), performing quality control, normalization, and dimensionality reduction with Scanpy and Squidpy
- Identified spatially variable genes using Moran's I and SpatialDE; performed unsupervised clustering to delineate cortical layer-specific gene expression domains and validated against histological annotations
- Built cell-type deconvolution pipeline using cell2location to estimate fine-grained cell-type proportions per spatial spot from paired single-cell RNA-seq reference data
- Visualized spatial gene expression patterns, cluster boundaries, and cell-type gradients overlaid on tissue morphology; reproduced key findings from the original publication

Knowledge and Skills: Python; Scanpy; Squidpy; cell2location; SpatialDE; spatial transcriptomics; 10x Visium; single-cell RNA-seq; dimensionality reduction; clustering; cell-type deconvolution; data visualization

Bulk RNA-seq Differential Expression Pipeline

Personal Project – GitHub

2025 – 2026

Berkeley, CA

- Built a reproducible Nextflow pipeline for bulk RNA-seq analysis: FastQC quality control, Trimmomatic adapter trimming, STAR alignment to GRCh38 reference genome, and featureCounts quantification
- Performed differential expression analysis using DESeq2 in R on a GEO dataset comparing tumor vs. normal tissue; identified 500+ significantly differentially expressed genes ($FDR < 0.05$, $|\log_2FC| > 1$)
- Conducted Gene Ontology and KEGG pathway enrichment analysis using clusterProfiler; identified upregulated pathways consistent with known tumor biology
- Generated publication-quality visualizations: volcano plots, MA plots, heatmaps of top DEGs, and PCA of sample-level variance; containerized pipeline with Docker for full reproducibility

Knowledge and Skills: Nextflow; Docker; FastQC; Trimmomatic; STAR; featureCounts; DESeq2; R; Python; clusterProfiler; KEGG; Gene Ontology; RNA-seq; differential expression; pathway enrichment; GEO; reproducible pipelines

PUBLICATIONS

Machine Learning-Based Phenotyping and Preoperative Prediction of Endometriosis Severity in Patients with Unexplained Infertility

Abstract submitted to Stanford AI in Medicine and Medical Imaging (AIMI) Conference. Manuscript in preparation for publication within Journal of Clinical Medicine.

Trichoscopic Oculomics: Integrating Machine Learning, Spatial Transcriptomics, and Metabolic Epidemiology to Reframe Androgenetic Alopecia as a Systemic Screening Opportunity

Manuscript in preparation for publication.

A Systematic Review of Home Video-Based Machine Learning Classification of Autism and ADHD in Pediatric Populations

Manuscript in preparation for publication.

25-min carotenoid blood test for lung cancer via a triple photonic resonance

Manuscript in preparation for publication.

TECHNICAL SKILLS

Bioinformatics & Omics: Scanpy, Squidpy, cell2location, SpatialDE, DESeq2, clusterProfiler, STAR, featureCounts, FastQC, Trimmomatic, KEGG, Gene Ontology, sequence alignment, genome analysis, primer design

Spatial Transcriptomics: 10x Visium, spatially variable gene detection, cell-type deconvolution, spatial clustering, tissue domain identification

Programming: Python, R, SQL, C++, MATLAB, Bash

Machine Learning: scikit-learn, XGBoost, TensorFlow, PyTorch, HuggingFace Transformers, LLMs, classification, regression, clustering, deep learning, CNNs, graph neural networks

Clinical NLP & Biomedical Text: BioBERT, named entity recognition, model cascades, benchmark design, MIMIC-III/IV, n2c2

Pipeline & Reproducibility: Nextflow, Docker, Git, Linux, AWS, CI/CD

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, statistical modeling

Molecular Biology: PCR, Gibson Assembly, yeast transformation, Cre-Lox, gel electrophoresis, Sanger sequencing, CRISPR, plasmid cloning

Synthetic Biology: Metabolic pathway engineering, gene cassette design, heterologous expression, recombinant DNA